

## **Unit 10: Clustering Algorithms and Cluster Analysis**

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### **Objectives**

After this lecture, you will be able to

- Learn the various measures of similarity.
- Know the concept of unsupervised learning.
- Understand the working of the K-Means Algorithm.
- Understand the working of K-medoids and Clarans clustering algorithms.
- Understand the working of various hierarchal clustering algorithms.
- Learn the concept of cluster analysis.
- Understand the concept of outliers and the different types of outliers.

### **Introduction**

Clustering is the process of grouping objects that are identical to one another. It can be used to determine if two things have similar or dissimilar properties. Clustering aids in the division of data into subsets. The data in each of these subsets is comparable, and these subsets are referred to as clusters. We can make an informed conclusion on who we believe is most suited for this product now that the data from our consumer base has been separated into clusters.

Outliers are cases that are out of the ordinary because they fall outside of the data's normal distribution. The distance from a normal distribution center reflects how typical a given point is

concerning the data's distribution. Each situation can be categorized as normal or atypical based on the likelihood of being typical or atypical.

## 10.1 Measures of Similarity

The similarity metric is a distance between two objects with dimensions that describe their characteristics. That is, if the distance between two data points is small, the objects have a high degree of similarity, and vice versa. The similarity is a subjective concept that is highly influenced by context and application.



The taste, size, color, and other characteristics of vegetables, for example, can be used to determine their similarity.

To evaluate the similarities or differences between two objects, most clustering methods use distance measures. The most commonly used distance measures are:

### Euclidean Distance

The standard metric for solving geometry problems is Euclidean distance. It is the ordinary distance between two points, to put it simply. It is one of the most widely used cluster analysis algorithms. K-mean is one of the algorithms that use this formula. Mathematically it computes the root of squared differences between the coordinates between two objects.

$$d(\mathbf{p}, \mathbf{q}) = d(\mathbf{q}, \mathbf{p}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (q_n - p_n)^2}$$

$$= \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Figure 1: Formula for Euclidean Distance

### Manhattan Distance

This defines the absolute difference between two coordinate pairs. To evaluate the distance between two points P and Q, simply measure the perpendicular distance between the points from the X-Axis and the Y-Axis. In a plane with P at coordinate (x1, y1) and Q at (x2, y2).

Manhattan distance between P and Q =  $|x_1 - x_2| + |y_1 - y_2|$

### Minkowski Distance

It's a combination of the Euclidean and Manhattan Distance Measurements. A point in an N-dimensional space is defined as (x1, x2, ..., xN)

Consider the following two points, P1 and P2:

**P1:** (X1, X2, ..., XN)

**P2:** (Y1, Y2, ..., YN)

The Minkowski distance between P1 and P2 is then calculated as follows:

$$\sqrt[p]{(x_1 - y_1)^p + (x_2 - y_2)^p + \dots + (x_N - y_N)^p}$$

Figure 2: Minkowski Distance Formula

- When  $p = 2$ , Minkowski distance is the same as the **Euclidean** distance.
- When  $p = 1$ , Minkowski distance is the same as the **Manhattan** distance.

| Attribute Type    | Dissimilarity                                                                                         | Similarity                                                                      |
|-------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Nominal           | $d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$                       | $s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$ |
| Ordinal           | $d = \frac{ p-q }{n-1}$<br>(values mapped to integers 0 to $n-1$ , where $n$ is the number of values) | $s = 1 - \frac{ p-q }{n-1}$                                                     |
| Interval or Ratio | $d =  p - q $                                                                                         | $s = -d, s = \frac{1}{1+d}$ or<br>$s = 1 - \frac{d - \min d}{\max d - \min d}$  |

Figure 3: Similarity/Dissimilarity for Simple Attributes

Where  $p$  and  $q$  are the attribute values for two data objects.

### Common Properties of a Distance

Distances, such as the Euclidean distance, have some well-known properties.

1.  $d(p, q) \geq 0$  for all  $p$  and  $q$  and  $d(p, q) = 0$  only if  $p = q$ . (Positive definiteness)
2.  $d(p, q) = d(q, p)$  for all  $p$  and  $q$ . (Symmetry)
3.  $d(p, r) \leq d(p, q) + d(q, r)$  for all points  $p, q$ , and  $r$ . (Triangle Inequality)

where  $d(p, q)$  is the distance (dissimilarity) between points (data objects),  $p$  and  $q$ . A distance that satisfies these properties is a metric.

### Common Properties of a Similarity

Similarities, also have some well-known properties.

1.  $s(p, q) = 1$  (or maximum similarity) only if  $p = q$ .
2.  $s(p, q) = s(q, p)$  for all  $p$  and  $q$ . (Symmetry)

where  $s(p, q)$  is the similarity between points (data objects),  $p$  and  $q$ .

## 10.2 Unsupervised Learning

Unsupervised learning is a type of machine learning in which models are trained using an unlabeled dataset and are allowed to act on that data without any supervision. Unsupervised learning is the process of training a computer using data that hasn't been categorized or named, and then allowing the algorithm to operate on that data without supervision. The machine's job here is to sort unsorted data into groups based on similarities, patterns, and differences without any previous data training.

Unlike supervised learning, there is no instructor present, which means the computer will not be trained. As a result, the computer is limited in its ability to discover hidden structures in unlabeled data on its own.

The goal of unsupervised learning is to:

- find the underlying structure of the dataset
- the group that data according to similarities
- represent that dataset in a compressed format.

There are two types of algorithms for unsupervised learning:

- **Clustering:** A clustering problem is one in which you want to find the data's underlying groupings, such as grouping customers based on their shopping habits.
- **Association:** People who buy  $X$  also want to buy  $Y$ , for example, is an example of an association rule learning problem in which you want to discover rules that describe large portions of your data.

## Working of Unsupervised Learning

Working of unsupervised learning can be understood by the below diagram:

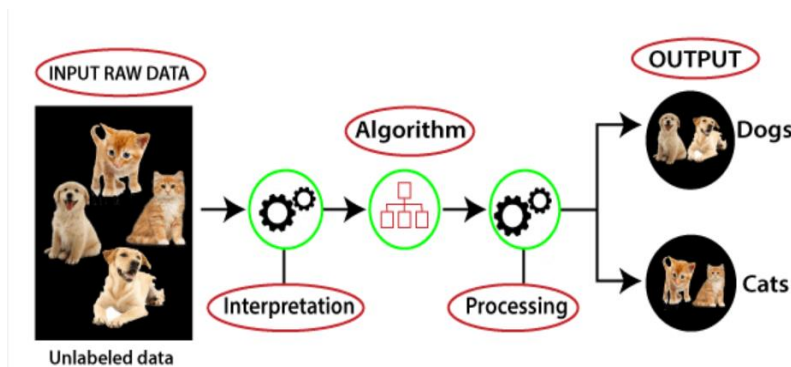


Figure 4: Unsupervised Learning

Assume it is shown a picture of both dogs and cats that it has never seen before. As a result, the machine has no understanding of the characteristics of dogs and cats, and we are unable to classify it as such. However, it can classify them based on their similarities, patterns, and differences, allowing us to easily divide the above image into two parts. The first part may contain all photos of dogs, while the second part may contain all photos of cats. It allows the model to work on its own to discover patterns and information that was previously undetected. It mainly deals with unlabelled data.



Recommender systems, which involve grouping together users with similar viewing patterns to recommend similar content.

### 10.3 K-Means Clustering

K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning. It is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, as if  $K=2$ , there will be two clusters, and for  $K=3$ , there will be three clusters, and so on. It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm. The k-means clustering algorithm mainly performs two tasks:

- Determines the best value for K center points or centroids by an iterative process.
- Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.

#### Working of K-Means Algorithm

Step-1: Select the number K to decide the number of clusters. Suppose we have two variables M1 and M2. The x-y axis scatter plot

of these two variables is given below:

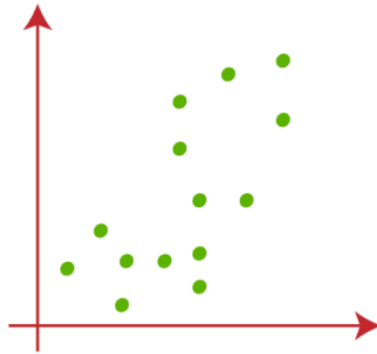


Figure 5: Scatter plot of M1 and M2

Step-2: Select random K points or centroids. It can be other from the input dataset. We need to choose some random k points or centroid to form the cluster. These points can be either the points from the dataset or any other point.

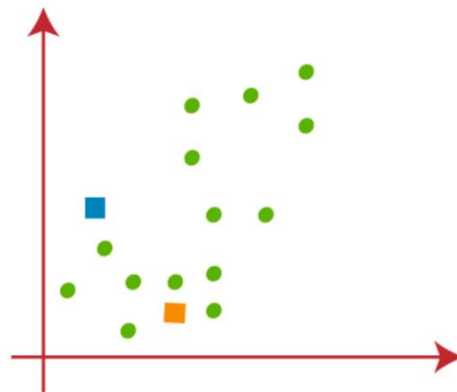


Figure 6: Selection of Centroid

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters. Now we will assign each data point of the scatter plot to its closest K-point or centroid. So, we will draw a median between both the centroids

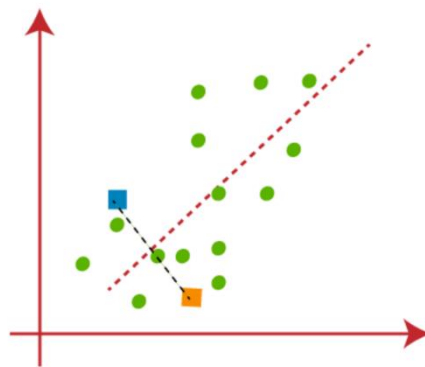


Figure 7: Assigning of data points to their closest centroid

Step-4: Calculate the variance and place a new centroid of each cluster. From the previous image, it is clear that points left side of the line are near to the blue centroid, and points to the right of the line are close to the yellow centroid. Let's color them as blue and yellow for clear visualization.

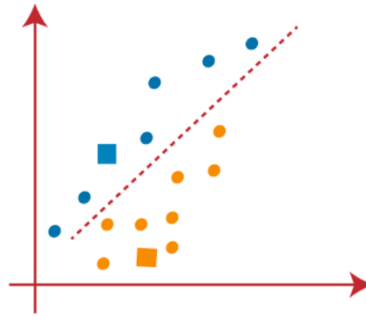


Figure 8: New centroid based upon new variance

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster. As we need to find the closest cluster, so we will repeat the process by choosing a **new centroid**. To choose the new centroids, we will compute the center of gravity of these centroids and will find new centroids

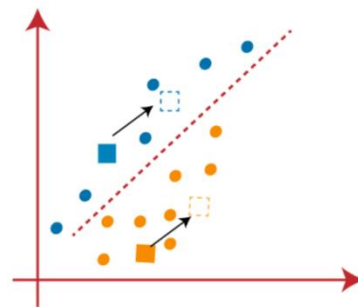


Figure 9: Reassign each point to the new closest centroid

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH. From the above image, we can see, one yellow point is on the left side of the line, and two blue points are right to the line. So, these three points will be assigned to new.

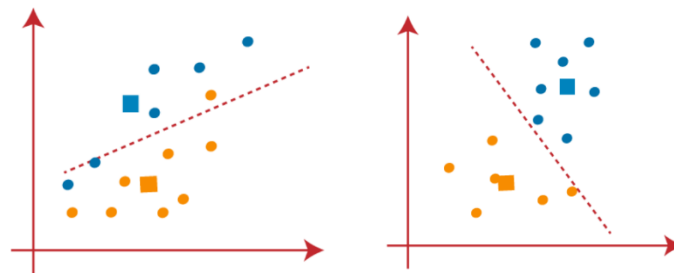


Figure 10: Points after reassignment

Step-7: The model is ready. As we got the new centroids so again will draw the median line and reassign the data points. As our model is ready, so we can now remove the assumed centroids and the two final clusters

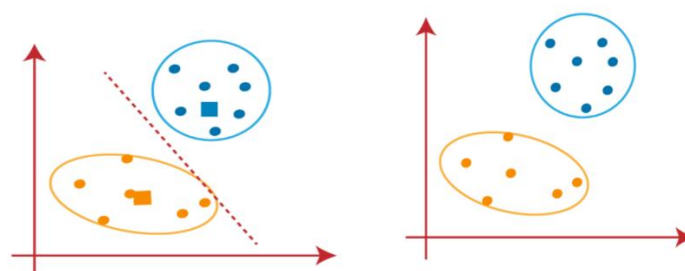


Figure 11: Final clusters

### 10.4 K-Medoids clustering(PAM)

A medoid is a point in a cluster whose dissimilarities with all other points in the cluster is its minimum.

$E = |P_i - C_i|$  is used to measure the dissimilarity of the medoid( $C_i$ ) and object( $P_i$ ).

In the K-Medoids algorithm, the cost is given as:

#### Algorithm

1. Initialize select k random points out of the n data points as the medoids.
2. Associate each data point to the closest medoid by using any common distance metric methods.
3. While the cost decreases:

For each medoid m, for each data o point which is not a medoid:

1. Swap m and o, associate each data point to the closest medoid, recompute the cost.
2. If the total cost is more than that in the previous step, undo the swap.

Let's consider the following example:

|   | X | Y |
|---|---|---|
| 0 | 8 | 7 |
| 1 | 3 | 7 |
| 2 | 4 | 9 |
| 3 | 9 | 6 |
| 4 | 8 | 5 |
| 5 | 5 | 8 |
| 6 | 7 | 3 |
| 7 | 8 | 4 |
| 8 | 7 | 5 |
| 9 | 4 | 5 |

Figure 12: Random Selection of Medoids

Step 1: Let the randomly chosen two medoids be C1 -(4, 5) and C2 -(8, 5) respectively.

Step 2: Calculating cost. Each non-medoid point's dissimilarity to the medoids is measured and tabulated:

|   | X | Y | Dissimilarity from C1 | Dissimilarity from C2 |
|---|---|---|-----------------------|-----------------------|
| 0 | 8 | 7 | 6                     | 2                     |
| 1 | 3 | 7 | 3                     | 7                     |
| 2 | 4 | 9 | 4                     | 8                     |
| 3 | 9 | 6 | 6                     | 2                     |
| 4 | 8 | 5 | -                     | -                     |
| 5 | 5 | 8 | 4                     | 6                     |
| 6 | 7 | 3 | 5                     | 3                     |
| 7 | 8 | 4 | 5                     | 1                     |
| 8 | 7 | 5 | 3                     | 1                     |
| 9 | 4 | 5 | -                     | -                     |

Figure 13: Similarity and dissimilarity index based upon selected centroids

Each point is assigned to the medoid cluster with less dissimilarity. Cluster C1 is represented by points 1, 2, 5; cluster C2 is represented by the points 0, 3, 6, 7, 8.

The cost=  $(3 + 4 + 4) + (3 + 1 + 1 + 2 + 2 + 2 + 2) = 20$

Step 3: Pick one non-medoid point at random and recalculate the cost. Allow the point to be chosen at random (8, 4). Each non-medoid point's dissimilarity to the medoids – C1 (4, 5) and C2 (8, 4) – is calculated and tabulated.

|   | X | Y | Dissimilarity from C1 | Dissimilarity from C2 |
|---|---|---|-----------------------|-----------------------|
| 0 | 8 | 7 | 6                     | 3                     |
| 1 | 3 | 7 | 3                     | 8                     |
| 2 | 4 | 9 | 4                     | 9                     |
| 3 | 9 | 6 | 6                     | 3                     |
| 4 | 8 | 5 | 4                     | 1                     |
| 5 | 5 | 8 | 4                     | 7                     |
| 6 | 7 | 3 | 5                     | 2                     |
| 7 | 8 | 4 | -                     | -                     |
| 8 | 7 | 5 | 3                     | 2                     |
| 9 | 4 | 5 | -                     | -                     |

Figure 14: Randomly selected new centroids

Each point is assigned to that cluster whose dissimilarity is less. So, points 1, 2, 5 go to cluster C1 and 0, 3, 6, 7, 8 go to cluster C2. The New cost = (3 + 4 + 4) + (2 + 2 + 1 + 3 + 3) = 22  
Swap Cost = New Cost – Previous Cost = 22 – 20 and 2 > 0

As the swap cost is not less than zero, we undo the swap. Hence (3, 4) and (7, 4) are the final medoids. The clustering would be in the following way

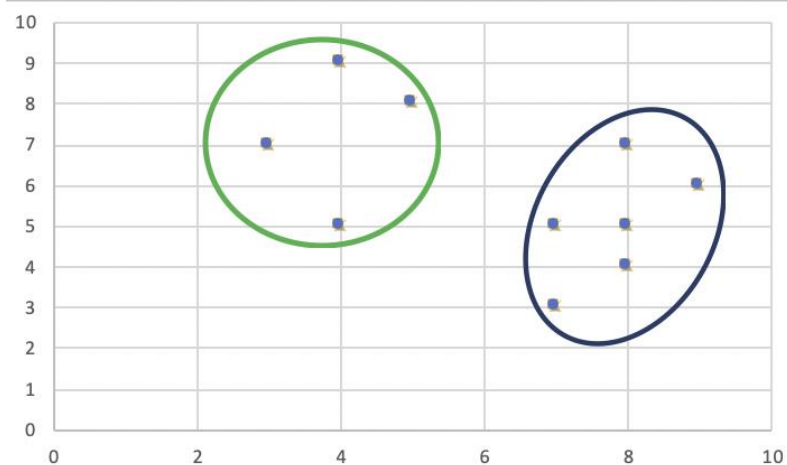


Figure 15: Cluster creation using K-Medoid

The time complexity is  $O(k * (n - k)^2)$ .

#### Advantages

1. The K-Medoid Algorithm is a fast algorithm that converges in a fixed number of steps.
2. It is simple to understand and easy to implement.
3. PAM is less sensitive to outliers than other partitioning algorithms.

#### Disadvantages

1. The key drawback of K-Medoid algorithms is that they cannot cluster non-spherical (arbitrary shaped) groups of points. This is because it relies on minimizing the distances between non-medoid objects and the medoid (cluster center) – in other words, it clusters based on compactness rather than connectivity.
2. Since the first k medoids are chosen at random, it can produce different results for different runs on the same dataset.



**Which method is more robust – k-means or k-medoids?**

The k-medoids method is more robust than k-means. Because in the presence of noise and outliers medoid is less influenced by outliers or other extreme values than a mean.

**10.5 The CLARANS algorithm**

Clustering Large Applications based upon RANdomized Search. It is an efficient medoid-based clustering algorithm. The k-medoids algorithm is an adaptation of the k-means algorithm.

CLARANS is a clustering partitioning method that is particularly useful in spatial data mining. By spatial data mining, we mean recognizing patterns and relationships in spatial data (such as distance-related, direction-related, or topological data, such as data plotted on a road map). CLARANS has two parameters:

**max neighbor**

- The maximum number of neighbors examined

**numerical**

- The number of local minima obtained.
- The higher the value of max neighbor, the closer is CLARANS to PAM

**Steps of CLARANS algorithm**

1. For the time being, select 'k' random data points and mark them as medoids.
2. Choose a random point, say 'a,' from the points chosen in step 1, as well as a point, say 'b,' that isn't included in those points.
3. Since that computation is needed for selecting the points in step 2, we would already have the number of distances of point 'a' from all other points (1). Carry out a similar calculation for point 'b.'
4. Replace 'a' with 'b' if the number of distances from all other points for point 'b' is less than that for point 'a'.
5. The algorithm runs a randomized search of medoids 'x' times, where 'x' is the number of local minima computed, i.e. the number of iterations to run, which we define as a parameter. The set of medoids obtained after such an 'x' number of measures is referred to as the **Local optimum**.
6. Any time substitution of points is made, a counter is incremented. The method of looking for potential replacement points is repeated until the counter reaches the maximum number of neighbors to be tested (specified as a parameter).
7. When the algorithm ends, the set of medoids obtained is the best local optimum option of medoids.

**CLARANS –Basic Idea**

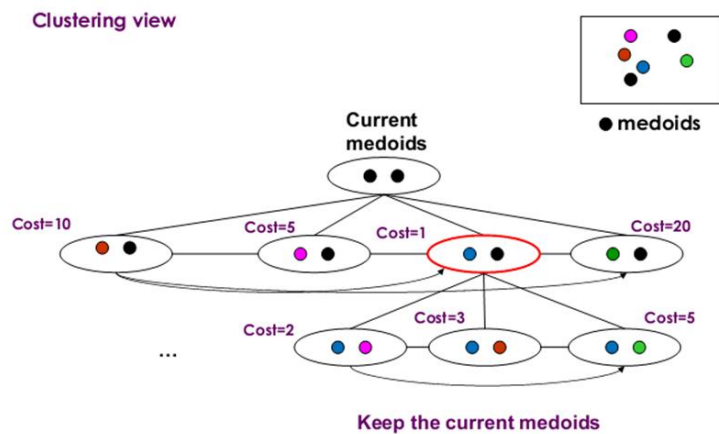


Figure 16: Selection of medoids

Draws a sample of nodes at the beginning of the search. Neighbors are from the chosen sample. Restricts the search to a specific area of the original data

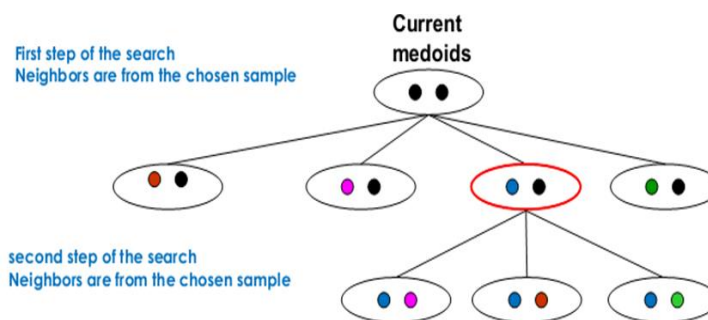


Figure 17: Bes local optimum.

Do not confine the search to a localized area. Stops the search when a local minimum is found. Finds several local optimums and output the clustering with the best local optimum.

#### Advantages

- Experiments show that CLARANS is more effective than both PAM and CLARA.
- Handles outliers

#### Disadvantages

- The computational complexity of CLARANS is  $O(n^2)$ , where  $n$  is the number of objects.
- The clustering quality depends on the sampling method.

## 10.6 Hierarchical Clustering

Hierarchical clustering begins by treating every data point as a separate cluster. Then, it repeatedly executes the subsequent steps:

- Identify the 2 clusters which can be closest together
- Merge the 2 maximum comparable clusters.
- We need to continue these steps until all the clusters are merged.

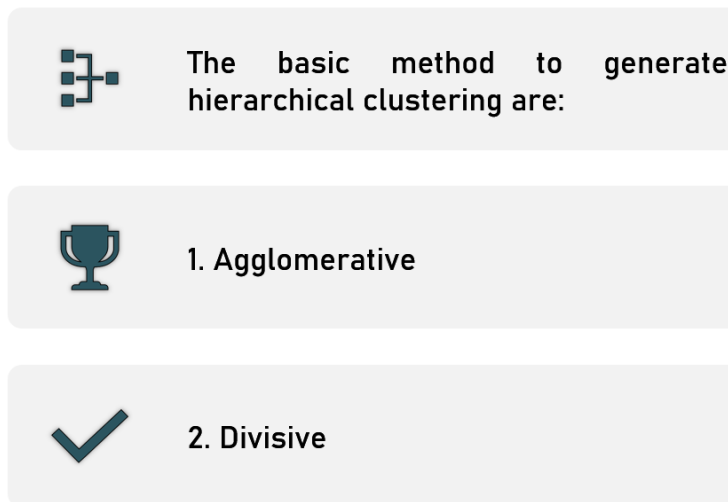


Figure 18: Types of Clustering algorithm

The basic method to generate hierarchical clustering are:

### 1. Agglomerative:

Consider and data point as a separate Cluster at first, then combine the cluster's nearest pairs at each stage. (It's a bottom-up approach.) At first, each data set is regarded as a separate entity or cluster. The clusters combine with other clusters in each iteration until only one cluster is formed.

Agglomerative Hierarchical Clustering Algorithm:

1. Calculate the degree of similarity between one cluster and all others (calculate proximity matrix)
2. Consider each data point as an individual cluster.
3. Combine clusters that are very similar or identical to one another.
4. For each cluster, recalculate the proximity matrix.
5. Steps 3 and 4 should be repeated until only one cluster remains.

Let's say we have six data points A, B, C, D, E, F.

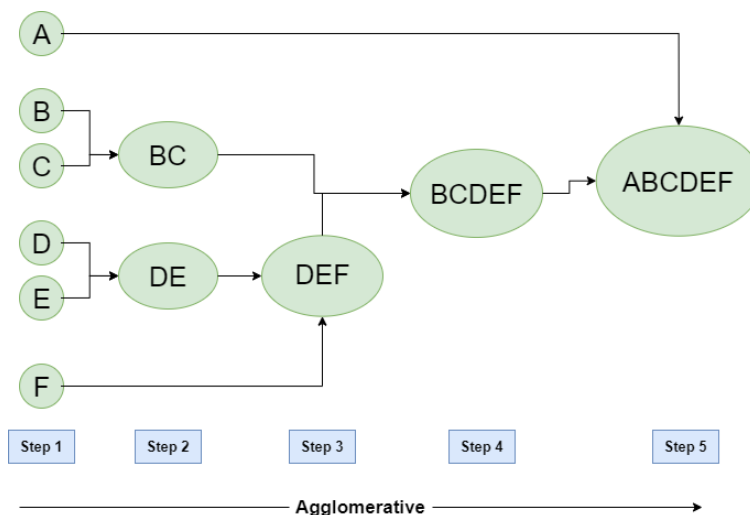


Figure 19: Agglomerative clustering

Step 1: Treat each alphabet as a separate cluster and measure the distance between each cluster and the others.

Step 2: Comparable clusters are combined to create a single cluster in the second step. Let's assume clusters (B) and (C) are very close, so we merge them in the second stage, just as we did with clusters (D) and (E), and we end up with clusters [(A), (BC), (DE), (F)].

Step 3: Using the algorithm, we recalculate the proximity and combine the two closest clusters  $[(DE), (F)]$  to form new clusters as  $[(A), (BC), (DEF)]$ .

Step 4: Repeat the process; the DEF and BC clusters are comparable and are combined to create a new cluster. Clusters  $[(A), (BCDEF)]$  are now all that's left.

Step 5: Finally, the remaining two clusters are combined to create a single cluster  $[(ABCDEF)]$ .

## 2. Divisive

Divisive Hierarchical Clustering is the exact opposite of Agglomerative Hierarchical Clustering. We consider all of the data points as a single cluster in Divisive Hierarchical clustering, and we distinguish the data points from the clusters that aren't equivalent in each iteration. We're left with N clusters in the end.

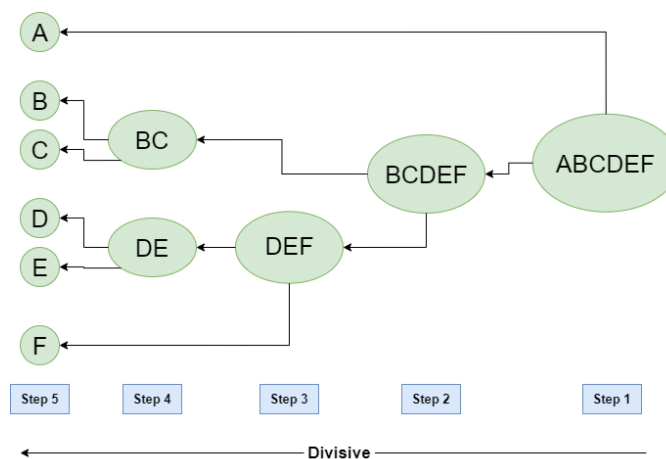


Figure 20: Divisive Clustering

## 10.7 BIRCH

BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies) is a clustering algorithm that clusters large datasets by first producing a small and compact summary of the large dataset that preserves as much information as possible. Instead of clustering the larger dataset, this smaller overview is clustered.

### Why BIRCH?

Clustering algorithms like K-means clustering do not perform clustering very efficiently and it is difficult to process large datasets with a limited amount of resources (like memory or a slower CPU). Regular clustering algorithms do not scale well in terms of running time and quality as the size of the dataset increases. This is where BIRCH clustering comes in.

Before we implement BIRCH, we must understand two important terms:

- **Clustering Feature (CF)**
- **CF - TreeClustering Feature (CF)**

**Clustering Feature:** BIRCH summarizes large datasets into smaller, dense regions called Clustering Feature (CF) entries. Formally, a Clustering Feature entry is defined as an ordered triple,  $(N, LS, SS)$ . A CF entry can be composed of other CF entries.

**CF Tree:** The CF tree is the compact representation we've been discussing up to this stage. A CF tree has a sub-cluster at each leaf node. A pointer to a child node and a CF entry made up of the number of CF entries in the child nodes are included in each entry in a CF tree. Each leaf node has a maximum number of entries. The threshold is the highest possible number.

### Parameters of BIRCH Algorithm

The following are the various parameters of the BIRCH Algorithm

**Threshold:** the threshold is the maximum number of data points a sub-cluster in the leaf node of the CF tree can hold.

**branching\_factor:** This parameter specifies the maximum number of CF sub-clusters in each node (internal node).

$n\_clusters$ : The number of clusters to be returned after the entire BIRCH algorithm is complete. If set to None, the final clustering step is not performed and intermediate clusters are returned.

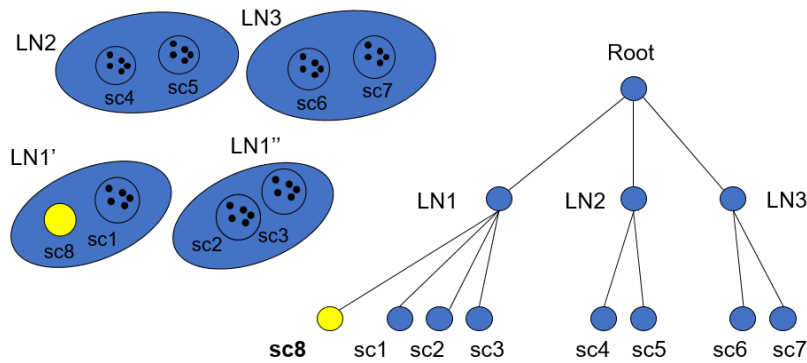


Figure 21: CF Tree Insertion

If the branching factor of a leaf node can not exceed 3 then the following tree results.

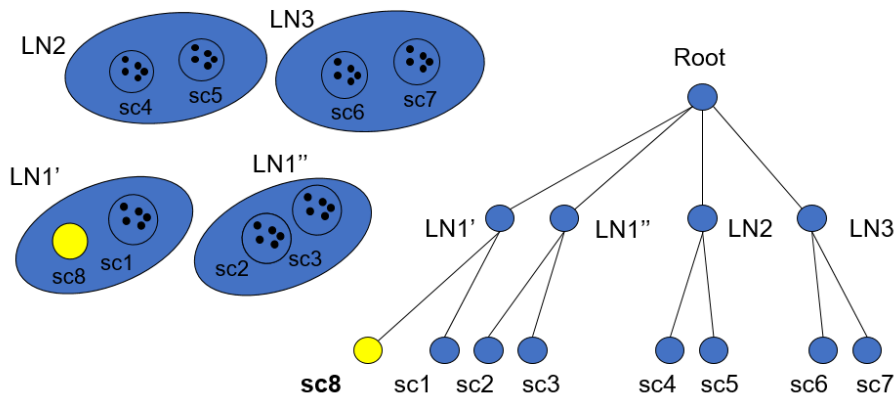


Figure 22: Leaf Branching Factor does not exceed 3

If the branching factor of a non-leaf node can not exceed 3, then the root is split, and the height of the CF Tree increases by one.

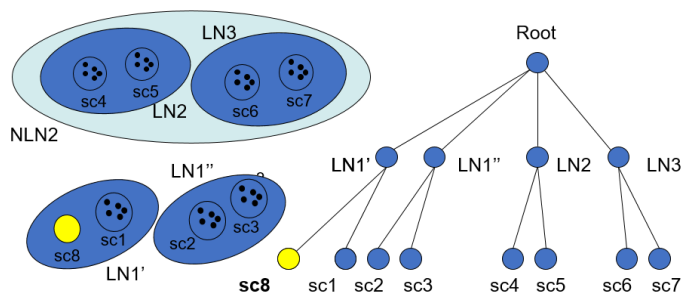


Figure 23: branching factor of a non-leaf node can not exceed 3

### Advantages

- Birch performs faster than existing algorithms (CLARANS and KMEANS) on large datasets.
- Scans whole data only once.
- Handles outliers better.

- Superior to other algorithms in stability and scalability.

#### Disadvantages

- Since each node in a CF tree can hold only a limited number of entries due to the size, a CF tree node doesn't always correspond to what a user may consider a nature cluster.
- Moreover, if the clusters are not spherical, it doesn't perform well because it uses the notion of radius or diameter to control the boundary of a cluster.

### 10.8 What is Cluster Analysis

The process of dividing a set of input data into possibly overlapping, subsets, where elements in each subset are considered related by some similarity measure.

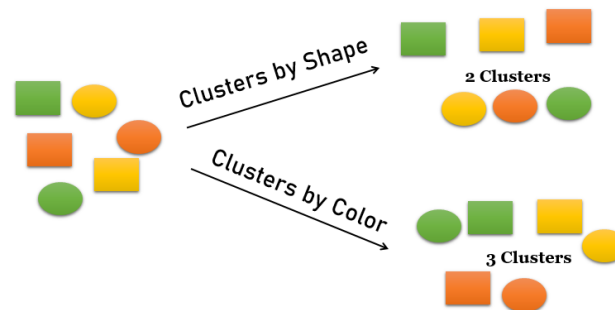


Figure 24: Clusters based upon shape and color

### Clustering Approaches

The following are the two clustering approaches that we are going to discuss:

- 1) DBSCAN
- 2) Graph-Based Clustering

#### DBSCAN

The density-based clustering algorithm has been extremely useful in identifying non-linear shape structures. The most commonly used density-based algorithm is DBSCAN (Density-Based Spatial Clustering of Applications with Noise). It makes use of the density reachability and density connectivity concepts.

**Density Reachability-** A point "p" is said to be density reachable from a point "q" if point "p" is within  $\epsilon$  distance from point "q" and "q" has a sufficient number of points in its neighbors which are within distance  $\epsilon$ .

**Density Connectivity** - A point "p" and "q" are said to be density connected if there exists a point "r" which has a sufficient number of points in its neighbors and both the points "p" and "q" is within the  $\epsilon$  distance. This is the chaining process. So, if "q" is neighbor of "r", "r" is neighbor of "s", "s" is a neighbor of "t" which in turn is neighbor of "p" implies that "q" is neighbor of "p".

#### Algorithm

The set of data points is  $X = x_1, x_2, x_3, \dots, x_n$ . Two parameters are needed by DBSCAN: (eps) and the minimum number of points required to form a cluster (minPts).

- 1) Begin at a random starting point that has never been visited before.
- 2) Using (All points within the distance are neighborhood), extract the neighborhood of this point.
- 3) If there are enough neighbors in the region, the clustering process begins, and the point is labeled as visited; otherwise, it is labeled as noise (Later this point can become part of the cluster).
- 4) If a point is found to be part of a cluster, its neighbors are also part of the cluster, and the process from step 2 is repeated for all neighborhood points. This process is repeated until all of the cluster's points have been calculated.
- 5) A new, previously unexplored point is retrieved and processed, resulting in the detection of a new cluster or noise.

6) Repeat this step until all of the points have been marked as visited.

### DBSCAN algorithm requires two parameters

- **Eps:** Maximum radius of the neighborhood.
- **MinPts:** Minimum number of points in an Eps-neighbourhood of that point.

In this algorithm, we have 3 types of data points.

- **core point** - which has at least min pts points in its neighborhood
- **border point** - one which has a core point in its neighborhood
- **noise point** - one which is neither a core nor a border point and is considered an outlier in the dataset

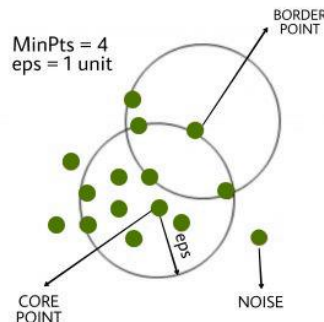


Figure 25: Points of DBSCAN

### Advantages

- 1) No need to specify the number of clusters ahead of time.
- 2) Capable of detecting noise data during clustering.
- 3) Clusters of any size and form can be found using the DBSCAN algorithm.

### Disadvantages

- 1) In the case of varying density clusters, the DBSCAN algorithm fails.
- 2) Fails in the case of datasets with a neck.
- 3) Doesn't work well for data with a lot of dimensions.

## 10.9 Graph-Based Clustering

Graph clustering refers to the clustering of data in the form of graphs. The following are the types of Graph Clustering:

- **Between-graph** - Clustering a set of graphs
- **Within-graph** - Clustering the nodes/edges of a single graph

**Between-Graph:** Between-graph clustering methods divide a set of graphs into different clusters.



A set of graphs representing chemical compounds can be grouped into clusters based on their structural similarity

**Within Graph:** Within-graph clustering methods divide the nodes of a graph into clusters.



In a social networking graph, these clusters could represent people with the same/similar hobbies

### K-Spanning Tree

Following are the steps to obtain a K-Spanning Tree:

- Obtains the Minimum Spanning Tree (MST) of input graph G

- Removes  $k-1$  edges from the MST
- Results in  $k$  clusters

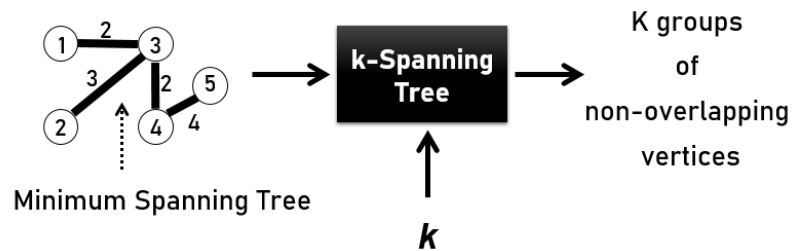


Figure 26: Working of K-Spanning tree

### What is the Minimum Spanning tree

A spanning tree of a connected and undirected graph is a subgraph that is a tree that links all of the vertices together. Many different spanning trees may exist in a single graph. For a weighted, connected, undirected graph, a minimum spanning tree (MST) or minimum weight spanning tree is a spanning tree with a weight less than or equal to the weight of any other spanning tree. The weight of a spanning tree is the sum of the weights assigned to each of its edges.

How many edges does a minimum spanning tree has? A minimum spanning tree has  $(V - 1)$  edges where  $V$  is the number of vertices in the given graph.

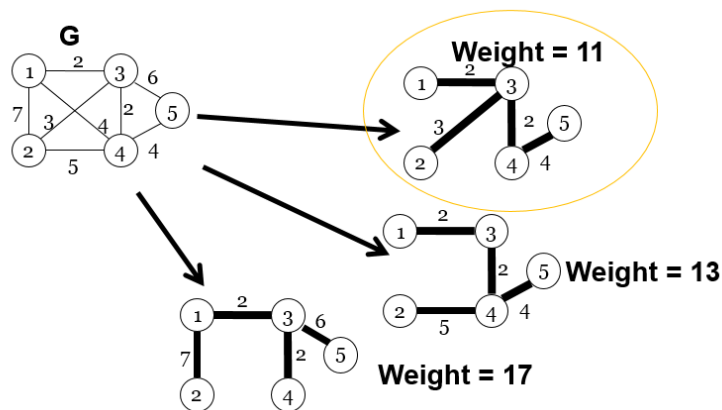


Figure 27: MST



maximum possible sum of edge weights, if the edge weights represent similarity

### Prim's Algorithm to Obtain MST

Given Input Graph G

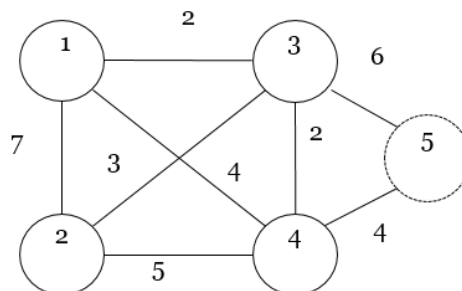


Figure 28: Graph



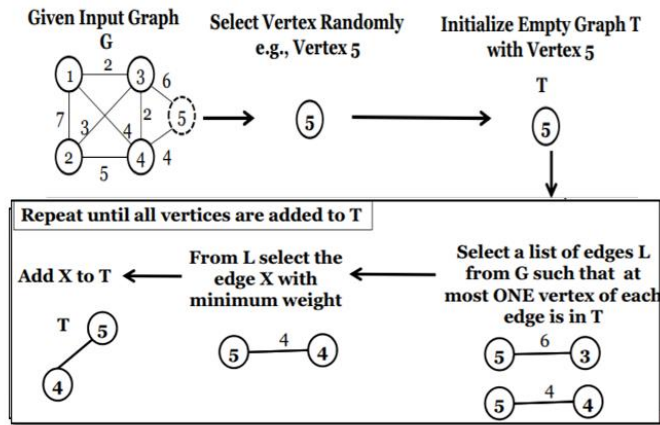


Figure 29: Steps for MST

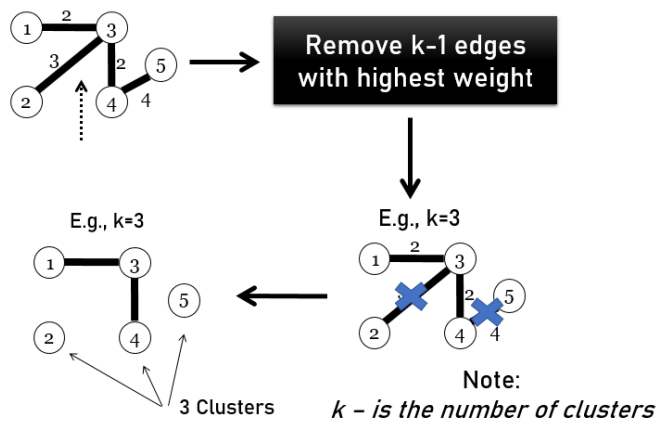


Figure 30: Formation of Cluster

### 10.10 Cluster Evaluation

Cluster evaluation assesses the feasibility of clustering analysis on a data set and the quality of the results generated by a clustering method.

Clustering evaluation includes the following tasks:

- Assessing clustering tendency.
- Determining the number of clusters in a data set.
- Measuring clustering quality

#### Assessing Clustering Tendency

Before applying any clustering method to your data, it's important to evaluate whether the data sets contain meaningful clusters (i.e.: non-random structures) or not. If yes, then how many clusters are there. This process is defined as the assessment of **clustering tendency**. The **Hopkins statistic** is used to assess the clustering tendency of a data set by measuring the probability that a given data set is generated by uniform data distribution. In other words, it tests the spatial randomness of the data.

The Hopkins statistic can be calculated as follow:

$$H = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i + \sum_{i=1}^n y_i}$$

A value for  $H$  higher than 0.75 indicates a clustering tendency at the 90% confidence level.

The null and the alternative hypotheses are defined as follow:

- **Null hypothesis:** the data set  $D$  is uniformly distributed (i.e., no meaningful clusters)
- **Alternative hypothesis:** the data set  $D$  is not uniformly distributed (i.e., contains meaningful clusters)

We can conduct the Hopkins Statistic test iteratively, using 0.5 as the threshold to reject the alternative hypothesis. That is, if  $H < 0.5$ , then it is unlikely that  $D$  has statistically significant clusters.

### Determining the Number of Clusters

Determining the “right” number of clusters in a data set is important, not only because some clustering algorithms like k-means require such a parameter, but also because the appropriate number of clusters controls the proper granularity of cluster analysis. It can be regarded as finding a good balance between compressibility and accuracy in cluster analysis.

What if you were to treat the entire data set as a cluster? This would maximize the compression of the data, but such a cluster analysis has no value.

On the other hand, treating each object in a data set as a cluster gives the finest clustering resolution. (i.e., most accurate due to the zero distance between an object and the corresponding cluster center). Figuring out the right number of clusters should often depend on the distribution's shape and scale in the data set, as well as the clustering resolution required by the user.



This approach of segmenting the database via clustering analysis is often used as an exploratory technique because it is not necessary for the end-user/analyst to identify ahead of time how records should be associated simultaneously.

### Measuring Clustering Quality

After clustering is complete, a variety of metrics can be used to assess how well the clustering worked. Minimal intra-cluster distance and maximum inter-cluster distance characterize ideal clustering. In general, these methods can be categorized into two groups according to whether ground truth is available or not. Ground Truth is factual data that has been observed or measured and can be analyzed objectively. It has not been inferred. If the data is based on an assumption, subject to opinion, or up for discussion, then, by definition, that is not Ground Truth data.

If ground truth is available, it can be used by **extrinsic methods**, which compare the clustering against the group truth and measure.

If the ground truth is unavailable, we can use **intrinsic methods**, which evaluate the goodness of clustering by considering how well the clusters are separated. Ground truth can be considered as supervision in the form of “**cluster labels**.” Hence, extrinsic methods are also known as supervised methods, while intrinsic methods are unsupervised methods. In general, a measure  $Q$  on clustering quality is effective if it satisfies the following four essential criteria:

**Cluster Homogeneity:** A clustering result satisfies homogeneity if all of its clusters contain only data points that are members of a single class.

**Cluster completeness:** A clustering result satisfies completeness if all the data points that are members of a given class are elements of the same cluster.

**Rag-Bag:** Elements with low relevance to the categories (e.g., noise) should be preferably assigned to the less homogeneous clusters (macro-scale, low-resolution, coarse-grained, or top-level clusters in a hierarchy).



How can I evaluate whether the clustering results are good or not when I try out a clustering method on the data set?

## 10.11 Outlier Detection and Analysis

An outlier is a data object that deviates significantly from the normal objects as if it were generated by a different mechanism. An outlier is a unique entity that stands out from the rest of the group.

They may be the result of a calculation or execution mistake. Outlier analysis, also known as outlier mining, is the study of outlier data.



Unusual credit card purchase, sports: Michael Jordon, Wayne Gretzky

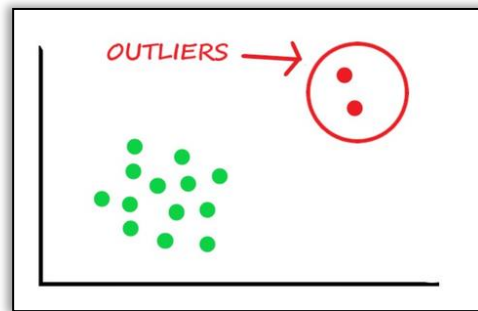


Figure 31: Outlier

### Why outlier analysis?

Most data mining methods ignore outliers, noise, or exceptions; however, in some applications, such as fraud detection, unusual events may be more interesting than those that occur more often, and thus outlier analysis becomes essential.

### Outlier Detection

To find the outlier, we must first set the threshold value such that every data point with a distance greater than it from its nearest cluster is considered an outlier for our purposes. The distance between the test data and each cluster means must then be determined. If the distance between the test data and the nearest cluster is greater than the threshold value, the test data will be classified as an outlier.

#### Algorithm for outlier detection

1. Calculate each cluster's average.
2. Initialize the Threshold value
3. Calculate the test data's distance from each cluster means.
4. Find the cluster that is closest to the test results.
5. Outlier exists if (Distance > Threshold).

### Challenges of Outlier Detection

#### 1. Modeling normal objects and outliers properly

- Hard to enumerate all possible normal behaviors in an application
- The border between normal and outlier objects is often a gray area

#### 2. Application-specific outlier detection

- Choice of distance measure among objects and the model of the relationship among objects are often application-dependent

E.g., clinic data: a small deviation could be an outlier; while in marketing analysis, larger fluctuations

#### 3. Handling noise in outlier detection

- Noise may distort the normal objects and blur the distinction between normal objects and outliers. It may help hide outliers and reduce the effectiveness of outlier detection

#### 4. Understandability

- Understand why these are outliers: Justification of the detection

- Specify the degree of an outlier: the unlikelihood of the object being generated by a normal mechanism

## Types of Outliers

The following are the different types of outliers:

- Global Outliers
- Contextual Outliers
- Collective Outliers.

### Global Outliers

A data point is considered a global outlier if its value is far outside the **entirety** of the data set in which it is found.

A **global outlier** is a measured sample point that has a very high or a very low value relative to all the values in a dataset. For example, if 9 out of 10 points have values between 20 and 30, but the 10th point has a value of 85, the 10th point may be a **global outlier**.



Figure 32: Global Outlier

### Contextual Outliers

If an individual data point is different in a specific context or condition (but not otherwise), then it is termed as a contextual outlier.

**Attributes of data objects should be divided into two groups:**

- Contextual attributes: defines the context, e.g., time & location
- Behavioral attributes: characteristics of the object, used in outlier evaluation, e.g., temperature

Contextual outliers are hard to spot if there was no background information. If you had no idea that the values were temperatures in summer, it may be considered a valid data point.

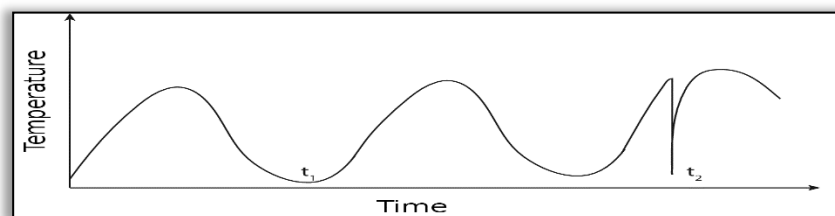


Figure 33: Contextual Outlier

### Collective Outliers

A subset of data objects *collectively* deviates significantly from the whole data set, even if the individual data objects may not be outliers. When several computers keep sending denial-of-service packages to each other.

Applications: E.g., *intrusion detection*:

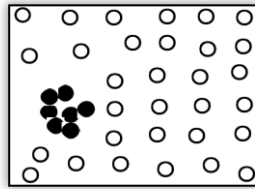


Figure 34: Collective Outliers



Give an example of a situation in which global outliers, contextual outliers, and collective outliers are all relevant. What are the characteristics, as well as the environmental and behavioral characteristics? In collective outlier detection, how is the link between items modeled?

### Summary

- **Types of outliers** include global outliers, contextual outliers, and collective outliers.
- An object may be more than one type of outlier.
- A **density-based method** cluster objects based on the notion of density. It grows clusters either according to the density of neighborhood objects (e.g., in DBSCAN) or according to a density function (e.g., in DENCLUE). OPTICS is a density-based method that generates an augmented ordering of the data's clustering structure.
- **Clustering evaluation** assesses the feasibility of clustering analysis on a data set and the quality of the results generated by a clustering method.
- A cluster of data objects can be treated as one group.
- Clustering analysis is broadly used in many applications such as market research, pattern recognition, data analysis, and image processing.
- **Outliers** are the **data** points that cannot be fitted in any type of cluster.
- The analysis of outlier data is referred to as outlier analysis or outlier mining.

### Keywords

**Clustering:** Clustering is a method of **data** analysis which groups **data** points to “maximizing the intraclass similarity and minimizing the interclass similarity.”

**Outliers:** outliers are data items that did not (or are thought not to have) come from the assumed population of data.

**Unsupervised learning:** This term refers to the collection of techniques where groupings of the data are defined without the use of a dependent variable. Cluster analysis is an example.

**Cluster Homogeneity:** A clustering result satisfies homogeneity if all of its clusters contain only data points that are members of a single class.

**Cluster Evaluation:** Cluster evaluation assesses the feasibility of clustering analysis on a data set and the quality of the results generated by a clustering method.

**Hopkins static:** Hopkins statistic is used to assess the clustering tendency of a data set by measuring the probability that a given data set is generated by uniform data distribution.

**Contextual attributes:** Attributes that define the context, e.g., time & location

### Self Assessment Questions

1. Which is needed by K-means clustering?
  - A. Defined distance metric
  - B. Number of clusters
  - C. Initial guess as to cluster centroids
  - D. All of these

2. K means and K-medoids are an example of which type of clustering method?

- A. Hierarchical
- B. Partition
- C. Probabilistic
- D. None of the above.

3. CLARANS stands for :

- A. Clustering Large Applications based upon RANdomized Search
- B. Clustering Long Applications based upon RANdomized Search
- C. Clustering Large Accounts based upon RANdomized Search
- D. Customize Large Applications based upon RANdomized Search

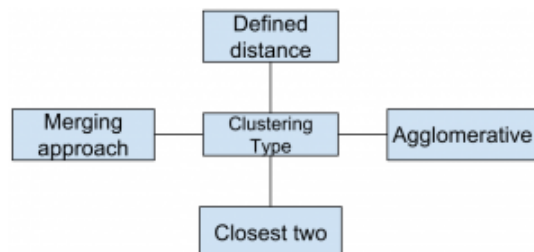
4. CLARANS combines sampling techniques with \_\_\_\_\_.

- A. K-means
- B. CLARA
- C. PAM
- D. All of the above

5. The k-medoids method is more robust than

- A. K-means
- B. CLARA
- C. PAM
- D. CLARANS

6. Which of the following clustering type has characteristics shown in the below figure?



- A. Partitional
- B. Hierarchical
- C. Naive Bayes
- D. None of the mentioned

7. Which of the following is finally produced by Hierarchical Clustering?

- A. Final estimate of cluster centroids
- B. Tree showing how close things are to each other
- C. Assignment of each point to clusters
- D. All of the mentioned

8. Which of the following clustering requires a merging approach?
- A. Partitional
  - B. Hierarchical
  - C. Naive Bayes
  - D. None of the mentioned
9. \_\_\_\_\_ specifies the maximum number of data points a sub-cluster in the leaf node of the CF tree can hold.
- A. n\_clusters
  - B. Branching\_factor
  - C. Threshold
  - D. All Of the above
10. Find the outliers in the given data set below.  
68, 6, 74, 70, 82
- A. 6
  - B. 82
  - C. 68
  - D. 70
11. Which statement about outliers is true?
- A. Outliers should be part of the training dataset but should not be present in the test data.
  - B. Outliers should be identified and removed from a dataset.
  - C. The nature of the problem determines how outliers are used
  - D. Outliers should be part of the test dataset but should not be present in the training data.
12. What does the term 'outlier' mean?
- A. A score that is left out of the analysis because of missing data
  - B. The arithmetic mean
  - C. A type of variable that cannot be quantified
  - D. An extreme value at either end of a distribution
13. Extreme values that occur infrequently are called as \_\_\_\_\_.
- A. Outliers.
  - B. Rare values.
  - C. Dimensionality reduction.
  - D. All of the above.
14. Which of the following is required by K-means clustering?
- A. Defined distance metric
  - B. Number of clusters

- C. Initial guess as to cluster centroids  
D. All of the mentioned

15. K-means is not deterministic and it also consists of several iterations.

- A. True  
B. False  
C. Can't Say  
D. Maybe

### **Answers for Self Assessment**

- |       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 1. D  | 2. B  | 3. A  | 4. C  | 5. A  |
| 6. B  | 7. B  | 8. B  | 9. B  | 10. A |
| 11. C | 12. D | 13. A | 14. D | 15. A |

### **Review Questions**

- Q1) Give an application example where global outliers, contextual outliers, and collective outliers are all interesting. What are the attributes, and what are the contextual and behavioral attributes?
- Q2) Briefly describe and give examples of each of the following approaches to clustering: partitioning methods, hierarchical methods, and density-based methods.
- Q3) For example explain the different types of outliers.
- Q4) Define hierarchical clustering along with its various types.
- Q5) Explain the K-mean algorithm.
- Q6) Differentiate between K-means and K-Medoid.
- Q7) Elucidate the step-by-step working of K-Medoid with example.

### **Further Reading**



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